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Correlations of Cerebral Indices for 'Extra' Cortical Parts and Ecological Variables in Primates

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Abstract. Correlations between cerebral indices for 'extra' cortical parts associated with advanced functions (Jerson's extra neurons, Nc; Hofman's extra cortical volume, Ve, and Hofman's extra cortical surface, Se) and ecological variables, namely troop size (TS), size of home range of one troop (HRt), and size of home range per individual (HRi) were examined in 12 species of prosimians, 8 species of New World monkeys, and 27 species of Old World monkeys. All three of the cerebral indices, Nc, Ve, and Se were, among the ecological variables examined, positively and most closely correlated with TS in the prosimians, with HRt in the New World monkeys, and with HRi in the Old World monkeys. These results suggest that the cerebral cortex may have expanded in relation to growth of TS for prosimians, in relation to expansion of HRt for New World monkeys, and in relation to expansion of HRi for Old World monkeys. It is, therefore, possible that the cerebral cortex may have expanded in conjunction with changes in different ecological factors in different taxonomic groups of primates.

Introduction

Although primates have developed various cerebral cortices, in particular the association cortices involved in higher functions, during the course of their evolution [Penfield, 1966; Sanides, 1970; Gross, 1973; Van Essen, 1979; Baker et al., 1981; Fuster, 1981; Allman, 1982; Hyvärinen, 1982; Kaas, 1983; Carlson et al., 1986], factors influencing the expansion or the development of the cerebral cortex of primates have not been clearly defined. It has been claimed that higher functions of primate brains, such as mnemonic, cognitive, and spatial functions, are adaptive features that evolved in response to the selection pressures exerted by various ecological factors [Jolly, 1972; Jerison, 1973, 1985; Humphrey, 1976; Cheney et al., 1986]. Since cerebral cortices are involved in such higher functions [Gross, 1973; Kubota, 1978; Fuster, 1981; Hyvärinen, 1982; Mishkin, 1982], the cerebral cortex in primates may have expanded in response to certain ecological factors. To identify such ecological factors, comparative studies of brains of living primates may prove useful. Indeed, recent studies have shown that 'comparative brain size' in living primates is related to diet and size of home range [Clutton-Brock and Harvey, 1980; Harvey et al., 1980] and suggested that the development of the brain may be associated with such ecological factors, although it is unknown whether the comparative brain size is indicative of the degree of the cortical expansion. In addition, it should be noted that the comparative brain size is significantly correlated with the size of home range for two families of Cercopithecidae and Pongidae, but is not for Lemuroidae, Lorisidae, and Cebidae [Clutton-Brock and Harvey, 1980]. This correlation suggests that ecological factors related to the development of the brain may differ across different taxonomic groups of primates. Although no definitive studies to examine whether the ecological factors that relate to the cortical expansion are the same or different across different taxonomic groups of primates have been carried out, such studies may pro-

vide evidence useful in our attempts to understand the nature of the evolution of the cerebral cortex in primates.

The aim of the present study is to reveal some of the ecological factors that are associated with cortical expansion in different taxonomic groups of primates and to investigate whether the same factors are related to cortical development in each primate group. With respect to the ecological factors, three ecological variables, namely, troop size (TS), the size of the troop's home range (HRt), and the size of the individual's home range (HRi) were chosen for the present study, since they are factors common to different taxonomic groups. The TS may be related to socialization or social interactions which have been suggested to be associated with the temporal and prefrontal cortices [Brody and Rosvold, 1952; Frazen and Myers, 1973]. HRt may be related to motoric, mnemonic, and spatial functions which are associated with various cerebral cortices, such as the parietal and temporal association cortices [Hyvärinen, 1982; Mishkin, 1982], and has been suggested to be associated with the development of the brain in primates [Clutton-Brock and Harvey, 1980; Harvey et al., 1980]. HRi has been shown to be closely related to quality of habitat [Takasaki, 1981, 1984]. Furthermore, to compare the degree of expansion or development of the cerebral cortex for several species of primates, appropriate quantitative indices for that are necessary, since the functional and anatomical organization of the cerebral cortex in several species of primates is not yet known. Some such indices have been calculated from body weight and brain weight, both of which are available for several species of primates. These indices have been based on various hypotheses, such as that of 'extra' cortical parts, such as Jerison's 'extra' neurons (Nc) [Jerison, 1955, 1973], and Hofman's 'extra' cortical volume (Ve) and surface (Se) [Hofman, 1982]. These hypotheses postulate, as general concepts, that mammalian cortex consists of a part associated with primitive functions (i.e., a putative brain part comparable to the brain of basal insectivores) and a part associated with 'advanced' functions. The indices of extra cortical parts, designated ECIs, are indicative of the size of parts of the cerebral cortex that are associated with advanced functions, and hence are indicative of the degree of cortical expansion. The ECIs have been shown to be closely related to learning ability or intelligence, as estimated from behavioral studies in several species of primates [Riddell et al., 1976; Riddell and Corl, 1977; Riddell, 1979; Ohta et al., 1984, 1985]. Arguing against Jerison's hypothesis of extra neurons, Holloway [1966, 1974] has pointed out that reorganization of the brain, rather than the addition of extra neurons, is essential during the evolution of primates. However, since the organization of the cerebral cortex for several species of primates is still unknown, these ECIs are useful, at least at present, for quantitative comparisons, between several species of primates, of the degree of cortical expansion, as well as of the size of cerebral cortices that are presumably associated with higher functions, such as association cortices. When the ECIs are significantly correlated with certain ecological variables for several species of primates, we can theorize that cortical expansion has been associated with such variables.

Thus, in the present study, correlations between ECIs and several ecological variables, namely, TS, HRt and HRi were examined in a number of species in three taxonomic groups: prosimians, New World monkeys, and Old World monkeys. The ecological variables correlated with the ECIs were different among the different taxonomic groups, suggesting that the expansion of the cerebral cortex may have been associated with different ecological factors among different taxonomic groups of primates.

Methods

Standard values of body weight and brain weight were obtained from Harvey and Clutton-Brock [1985]. When the body weight and/or brain weight differed between females and males in a given species, the mean value was adopted. Values of indices of ECIs, that is Nc, Ve and Se, were calculated from these values of brain weight and body weight, using formulae of Jerison [1973] and Hofman [1982] as follows (P = body weight; E = brain weight):

$$Nc = [E^{2/3}-(0.03P^{2/3})^{2/3}] \times 8 \times 10^{7},$$
$$Ve = 0.03P^{0.732}[(E/0.015P^{0.732})^{1.273}-1], \text{ and}$$
$$Se = 0.065P^{2/3}\{[E/(0.015P^{0.732})]-1\}.$$

The ecological variables used, TS, HRt, and HRi, were defined as follows: TS was the number of animals that form a troop and share a common home range; HRt was the total area in hectares used by a troop, and HRi was the total area of the home range divided by the TS. For monkeys that do not form troops (i.e. solitary prosimians), the values of HRi were equal to those of HRt. Standard values of these variables for primates were obtained from data obtained in several ecological field studies and cited by Milton and May [1976] and/or Clutton-Brock and Harvey [1977a, b].

Systematic data were available for 47 species, and these species were divided into three taxonomic groups: prosimians (Lemuroi-

Table I. Correlation coefficients and their levels of significance between $\log_{10}$ Nc, Ve and Se and $\log_{10}$ brain weight (E) and body weight (P) for prosimians, New World (NW) monkeys, and Old World (OW) monkeys

	Prosimians (n = 12)			NW monkeys (n = 8)			OW monkeys (n = 27)		
	Nc	Ve	Se	Nc	Ve	Se	Nc	Ve	Se
E	r = 0.98 ***	r = 0.99 ***	r = 0.99 ***	r = 0.99 ***	r = 0.96 ***	r = 0.97 ***	r = 0.97 ***	r = 0.96 ***	r = 0.98 ***
P	r = 0.90 ***	r = 0.92 ***	r = 0.93 ***	r = 0.79 *	r = 0.76 *	r = 0.79 *	r = 0.62 ***	r = 0.60 ***	r = 0.67 ***

* $p < 0.05$; *** $p < 0.001$ (*t test*). *n = Number of species examined.*

Table II. Correlation coefficients and their levels of significance between $\log_{10}$ Nc, Ve, Se brain weight (E), and body weight (P) and $\log_{10}$ TS, HRt, and size of HRi in prosimians, New World (NW) monkeys, and Old World (OW) monkeys

	Prosimians (n = 12)			NW monkeys (n = 8)			OW monkeys (n = 27)		
	TS	HRt	HRi	TS	HRt	HRi	TS	HRt	HRi
Nc	r = 0.71 **	r = 0.55 NS	r = 0.17 NS	r = 0.59 NS	r = 0.77 *	r = 0.56 NS	r = 0.32 NS	r = 0.74 ***	r = 0.81 ***
Ve	r = 0.71 **	r = 0.57 NS	r = 0.19 NS	r = 0.55 NS	r = 0.82 *	r = 0.64 NS	r = 0.30 NS	r = 0.74 ***	r = 0.82 ***
Se	r = 0.71 **	r = 0.58 NS	r = 0.19 NS	r = 0.55 NS	r = 0.81 *	r = 0.63 NS	r = 0.27 NS	r = 0.72 ***	r = 0.80 ***
E	r = 0.69 *	r = 0.61 *	r = 0.23 NS	r = 0.55 NS	r = 0.73 *	r = 0.56 NS	r = 0.21 NS	r = 0.66 ***	r = 0.74 ***
P	r = 0.66 *	r = 0.65 *	r = 0.29 NS	r = 0.28 NS	r = 0.38 NS	r = 0.42 NS	r = 0.06 NS	r = 0.29 NS	r = 0.37 NS

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$ (t test). n = Number of species examined; NS = not significant at 5% level.

dea and Lorisoidea, including 8 genera and 12 species); New World monkeys (Ceboidea, including 7 genera and 8 species); and Old World monkeys (Cercopithecoidea, including 10 genera and 27 species). These data are listed in the Appendix. All variables were normalized by logarithmic transformation.

Results

Correlations between Nc, Ve and Se (ECIs) and the ecological variables, TS, HRt, and HRi, were examined in three taxonomic groups: prosimians, New World monkeys, and Old World monkeys. Correlations of ECIs with brain weight and body weight were also examined. As shown in table I, all ECIs were significantly and positively correlated with brain weight and body weight. It was therefore possible that the correlations between ECIs and the ecological variables, which are discussed below, simply reflect results derived from the relationships between these weight and the ecological variables. To examine this possibility, correlations of brain weight with ecological variables, and correlations of body weight with the same variables were also examined. Furthermore, in addition to the single correlation analysis between each ECI and each ecological variable, multiple regression analysis was also performed in an attempt to examine which ecological variables were most closely correlated with each ECI for each group. Similar results were obtained for each ECI; results of analyses of correlations with Nc are, therefore, described in detail, while those with Ve and Se are described briefly.

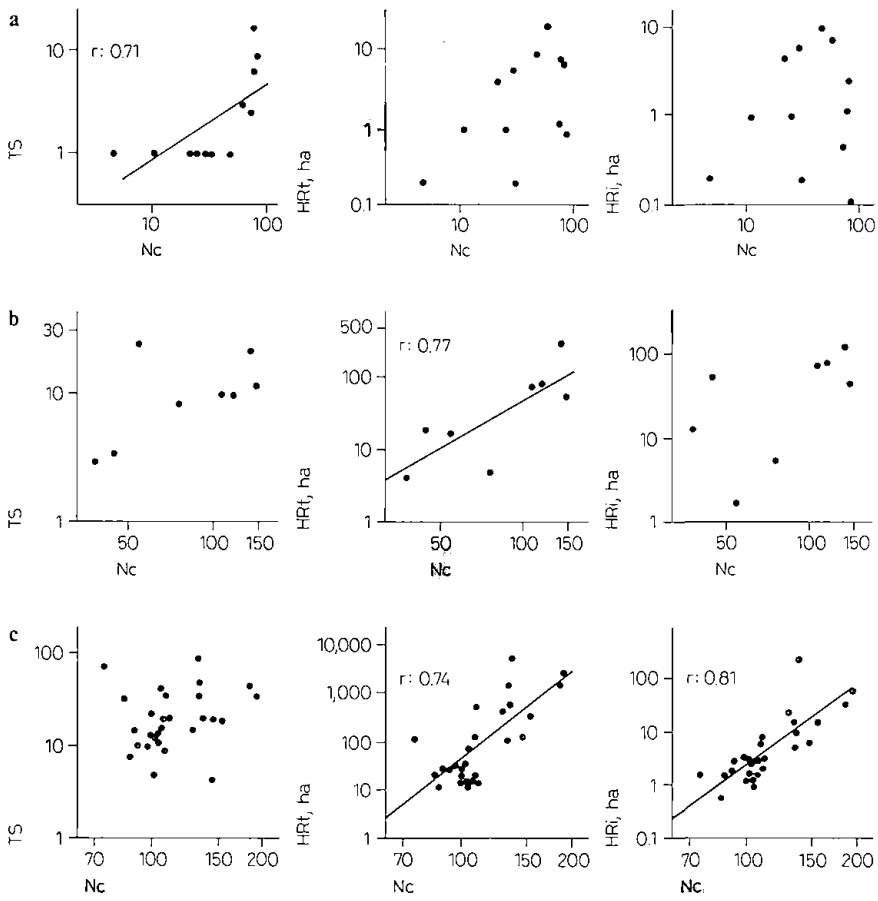


Fig. 1. Ecological variables plotted against Nc ($/10^7$) in prosimians **(a)**, in New World monkeys **(b)**, and in Old World monkeys **(c)**. Nc is correlated with TS in the prosimians, with HRt in the New World monkeys, and with HRt and HRi in the Old World monkeys. The equation for the regression line for the correlation of TS with Nc in the prosimians is log TS = $-0.73 + 1.20$ log $Nc/10^7$, that for the correlation of HRt with Nc in the New World monkeys is $\log_{10}$ HRt = $-2.78 + 2.25$ log $Nc/10^7$, that for the correlation of HRt with Nc in the Old World monkeys is $\log_{10}$ HRt = $-10.25 + 5.93$ log $Nc/10^7$, and that for the correlation of HRi with Nc in the Old World monkeys is $\log_{10}$ HRi = $-9.60 + 5.00$ log $Nc/10^7$. r = Correlation coefficient.

Correlations between Nc and Ecological Variables

Coefficients and levels of significance of correlations between Nc and the ecological variables in different taxonomic groups are shown in table II, and the ecological variables are plotted against Nc for each group in figure 1. As shown in table II and figure 1, the distribution of the ecological variables that correlated with Nc differed across the three taxonomic groups.

Prosimians. In the prosimians (n = 12), among TS, HRt, and HRi, Nc was significantly and positively correlated with TS (r = 0.71; t = 3.20; df = 10; $p < 0.01$), as shown in table II and figure 1a. There was a slight correlation between HRt and Nc, but the correlation did not reach the significance level of 5% (r = 0.55; t = 2.08; df = 10; $p < 0.1$; nonsignificant). No correlation was found between HRi and Nc (r = 0.17; t = 0.55; df = 10; nonsignificant) (table II). The solitary prosimians usually had smaller Nc than the group-forming prosimians (analysis of variance: F = 14.55; df = 1, 10; $p < 0.01$). When only the prosimians that form groups (5 species) were examined, there was a high positive coefficient (r = 0.76) of correlation between Nc and TS seen, although the correlation did not reach the level of statistical significance (t = 2.05; df = 3; $0.1 < p < 0.2$; nonsignificant). The equation for the regression line for describing correlation was:

$$\log_{10} Nc/10^7 = 1.57 + 0.07 \times \log_{10} TS.$$

To examine further which ecological variables were most closely correlated with Nc, multiple regression analysis [see Overall and Klett, 1973] was attempted. The best-fit multiple regression line for the multiple correlation of Nc and the ecological variables was given, with partial-regression coefficients, as follows:

$$\log_{10} Nc/10^7 = 1.25 + 0.42 \times \log_{10} TS.$$

When standard partial-regression coefficients were used, this equation became:

$$\log_{10} Nc/10^7 = 0.71 \times \log_{10} TS.$$

The multiple regression line represented by the latter equation was statistically significant at the 1% level (analysis of variance: df = 1, 10; F = 10.20). The multiple-correlation coefficient for this formula was 0.71, which was the same as the single-correlation coefficient for Nc and TS. These data indicate that, among TS, HRt, and HRi, TS was most closely correlated with Nc in the prosimians.

New World Monkeys. In the case of the New World monkeys (n = 8), Nc was positively correlated with HRt (r = 0.77; t = 3.20; df = 6; $p < 0.05$), but not with TS (r = 0.61; t = 1.88; df = 6; nonsignificant), or HRi (r = 0.57; t = 1.71; df = 6; nonsignificant) (table II, fig. 1b). The equation for the regression line of the correlation between Nc and HRt in the New World monkeys was:

$$\log_{10} Nc/10^7 = 1.52 + 0.26 \times \log_{10} HRt.$$

When multiple regression analysis was attempted, the best-fit multiple regression line was given, using partial-regression coefficients, as follows:

$$\log_{10} Nc/10^7 = 1.52 + 0.26 \times \log_{10} HRt.$$

With standard partial-regression coefficients, this equation beccame:

$$\log_{10} Nc/10^7 = 0.77 \times \log_{10} HRt.$$

The regression line represented by the latter equation was statistically significant at the 5% level (analysis of variance: df = 1, 6; F = 8.94). The multiple-correlation coefficient of 0.77 was the same as the single-correlation coefficient for Nc and HRt. These data indicate that, among the ecological variables examined, HRt was most closely correlated with Nc in the New World monkeys.

Old World Monkeys. In the Old World monkeys (n = 27), Nc was positively correlated with HRt (r = 0.74; t = 5.53; df = 25; $p < 0.001$) and HRi (r = 0.81; t = 6.79; df = 25; $p < 0.001$), but not with TS (r = 0.28; t = 1.46; df = 25; nonsignificant) (table II, fig. 1c). The coefficient of the correlation between Nc and HRi was larger than that between Nc and HRT (table II). The equation for the regression line of the correlation of Nc and HRi in the Old World monkeys was:

$$\log_{10} Nc/10^7 = 1.97 + 0.13 \times \log_{10} HRi,$$

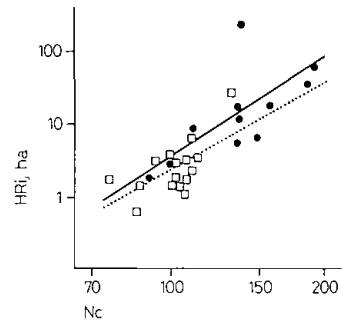


Fig. 2. The $\log_{10}$ HRi plotted against $\log_{10}$ $Nc/10^7$ for the terrestrial (●) and the arboreal species (□) of Old World monkey. The equation for the regression lines are $\log_{10}$ HRi = −7.96 + 4.27 $\log_{10}$ $Nc/10^7$ for the terrestrial species (——; r = 0.69; $p < 0.05$) and $\log_{10}$ HRi = −8.18 + 4.27 $\log_{10}$ $Nc/10^7$ for the arboreal species (·····; r = 0.63; $p < 0.01$).

Table III. Correlation coefficients and their levels of significance between $\log_{10}$ size of HRi and $\log_{10}$ Nc, body weight (P) and brain weight (E), in the terrestrial and arboreal species of Old World monkeys

	Terrestrial (n = 11)	Arboreal (n = 16)
NC	r = 0.69	r = 0.63
	*	**
E	r = 0.66	r = 0.46
	*	NS
P	r = 0.54	r = 0.07
	NS	NS

* $p < 0.05$; ** $p < 0.01$ (t test). n = Number of species examined; NS = not significant at 5% level.

and that of Nc and HRt was:

$$\log_{10} Nc/10^7 = 1.87 + 0.09 \times \log_{10} HRt.$$

As data for several species were available among the genera *Presbytis*, *Macaca* and *Cercopithecus* of the Old World monkeys, the intrageneric correlation between Nc and HRi was examined in these genera. There was a significant and positive correlation between Nc and HRi in *Presbytis* (r = 0.92; t = 4.69; df = 4; $p < 0.01$). A high and positive coefficient of correlation was found between Nc and HRi in *Macaca* (r = 0.79), although the correlation did not reach the level of significance of 5% (t = 2.54; df = 4; $p < 0.1$; nonsignificant). In *Cercopithecus*, no significant correlation was found between Nc and HRi (r = 0.20; t =

Table IV. Results of multiple-regression analysis of correlations of $\log_{10}$ Nc, Se, and Ve with $\log_{10}$ TS, HRt, and HRi in prosimians, New World (NW) monkeys, and Old World (OW) monkeys

	Best-fit regression line	r	F	df
Prosimians				
Nc	$\log Nc = 1.25 + 0.42 \times \log TS$	0.71**	10.20	1, 10
Ve	$\log Ve = 0.30 + 0.60 \times \log TS$	0.71**	10.18	1, 10
Se	$\log Se = 1.22 + 0.55 \times \log TS$	0.71**	10.12	1, 10
NW monkeys				
Nc	$\log Nc = 1.52 + 0.26 \times \log HRt$	0.77*	8.94	1, 6
Ve	$\log Ve = 0.65 + 0.41 \times \log HRt$	0.82*	10.66	1, 6
Se	$\log Se = 1.54 + 0.36 \times \log HRt$	0.81*	10.22	1, 6
OW monkeys				
Nc	$\log Nc = 1.97 + 0.13 \times \log HRi$	0.81***	48.38	1, 25
Ve	$\log Ve = 1.31 + 0.19 \times \log HRi$	0.82***	50.29	1, 25
Se	$\log Se = 2.14 + 0.17 \times \log HRi$	0.80***	44.38	1, 25

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$ (analysis of variance). r = Standard partial-regression coefficient; F = ratio of variance; df = degree of freedom.

0.40; df = 4; nonsignificant). This absence of a significant correlation may be due to the fact that values of Nc and HRi for species of *Cercopithecus* were not as varied as those for species of *Presbytis* and *Macaca*.

Since a considerable number of species of Old World monkeys are terrestrial, being dissimilar from prosimians and New World monkeys which are mostly arboreal, and since HRi is related to the quality of habitat [Takasaki, 1981, 1984], which may differ between the arboreal and terrestrial habitats, correlations between Nc and HRi, and the distribution of their values were examined separately for arboreal (n = 16) and terrestrial species (n = 11). For both the terrestrial and arboreal species, Nc was significantly correlated with HRi ($t = 2.85$; df = 9; $p < 0.05$, for the terrestrial species; $t = 3.06$; df = 14; $p < 0.01$, for the arboreal species), as shown in figure 2 and table III. The equation for the regression lines were

$$\log_{10} Nc/10^7 = 2.00 + 0.11 \times \log_{10} HRi$$

for the terrestrial species and

$$\log_{10} Nc/10^7 = 1.97 + 0.09 \times \log_{10} HRi$$

for the arboreal species. The correlation coefficient was 0.69 for the terrestrial species and 0.63 for the arboreal species, and these coefficients were not significantly different from each other ($z = 0.24$; $p > 0.5$). Furthermore, the values of Nc for the terrestrial species were significantly larger than those for the arboreal species (analysis of variance: $F = 18.36$; df = 1, 25; $p < 0.001$), as well as the values of HRi ($F = 16.5$; $p < 0.001$). Thus, Nc was correlated with HRi for both the terrestrial and arboreal species, and the terrestrial species tended to have larger values of Nc and wider areas of HRi than the arboreal species.

To examine further the association between Nc and the ecological variables for the Old World monkeys, multiple regression analysis was attempted. The equation for the best-fit multiple regression line, using partial-regression coefficients, was as follows:

$$\log_{10} Nc/10^7 = 1.97 + 0.13 \times \log_{10} HRi.$$

With standard partial-regression coefficients, this equation became:

$$\log_{10} Nc/10^7 = 0.81 \times \log_{10} HRi.$$

The regression line represented by the latter equation was statistically significant at the 0.1% level (analysis of variance: df = 1, 25; $F = 48.38$). The multiple-correlation coefficient of 0.81 was the same as the

single-correlation coefficient for Nc and HRi. These data indicate that, among the ecological variables examined, HRi was most closely correlated with Nc for the Old World monkeys.

Correlations of Ve and Se with Ecological Variables

Distributions of the ecological variables correlated with Ve were the same as those correlated with Nc, as shown in table II. Ve was significantly correlated with TS in the prosimians, with HRt in the New World monkeys, and with HRt and HRi in the Old World monkeys. Coefficients of the correlations and their levels of significance for Ve and the ecological variables were similar to those for Nc and the ecological variables. As for Se, the distributions of ecological variables that were correlated with Se were the same as those in the case of Ve and Nc, as shown in table II. Furthermore, correlation coefficients and their levels of significance for Se were similar to those for Ve and Nc (table II). Moreover, as shown in table IV, the multiple regression analysis provided similar results for each ECI of Nc, Se and Ve: among the ecological variables examined, each ECI was most closely correlated with TS in the case of the prosimians, with HRt in the case of the New World monkeys, and with HRi in the case of the Old World monkeys. Thus, all ECIs generated similar results.

Correlations between Brain Weight and Ecological Variables

Correlations between brain weight and ecological variables are shown in table II. In the prosimians examined, brain weight was positively correlated with TS ($r = 0.69$; $t = 3.09$; $df = 10$; $p < 0.05$) and HRt ($r = 0.61$; $t = 2.46$; $df = 10$; $p < 0.05$), but not with HRi ($r = 0.23$; $t = 0.76$; $df = 10$; nonsignificant). The coefficient of correlation of brain weight with TS was larger than that of brain weight with HRt. In the New World monkeys, brain weight was correlated with only HRt ($r = 0.73$; $t = 2.60$; $df = 6$; $p < 0.05$) (table II). In the Old World monkeys, brain weight was correlated with HRt ($r = 0.66$; $t = 4.36$; $df = 25$; $p < 0.001$) and HRi ($r = 0.74$; $t = 5.46$; $df = 25$; $p < 0.001$), and the correlation coefficient between brain weight and HRi was larger than that between brain weight and HRt (table II). Thus, the distribution of the ecological variables correlated with brain weight, across the three taxonomic groups, was similar to the distribution of ecological variables correlated with the ECIs. However, coefficients of correlations with brain weight were usually lower than those with any of the ECIs, a result which indicates that correlations between ECIs and the ecological variables are stronger than correlations between those variables and brain weight.

With respect to the correlation between brain weight and HRi for the arboreal and terrestrial species of Old World monkey, the terrestrial species showed significant and positive correlations between these two parameters, but the arboreal species did not (table III). For the terrestrial species, the coefficient of correlation between brain weight and HRi ($r = 0.66$) was somewhat lower than that between Nc and HRi ($r = 0.69$), as shown in table III. This difference indicates that the correlation between Nc and HRi for the terrestrial and arboreal species of Old World monkey, as described above, is stronger than that between brain weight and HRi of these monkeys.

Correlations between Body Weight and Ecological Variables

Correlations of body weight with the ecological variables are shown in table II. In the prosimians, there was a positive correlation between body weight and TS ($r = 0.66$; $t = 2.75$; $p < 0.05$). This correlation coefficient was lower than that between any ECI and TS. In the New World monkeys and in the Old World monkeys, no correlations were found between body weight and the ecological variables (table II). Thus, the correlations of ECIs and the ecological variables were not simply a result of correlations between body weight and the ecological variables as tentatively postulated above.

With respect to correlation of body weight and HRi for arboreal and terrestrial species of Old World monkeys, neither the terrestrial nor the arboreal species showed any significant correlations between these parameters, as shown in table III. Furthermore, there was no significant difference in body weight between the arboreal and terrestrial species ($F = 1.45$; $df = 1, 25$; $p > 0.1$). Thus, the above-mentioned significant correlation between Nc and HRi for the arboreal and terrestrial species of Old World monkeys was not simply the result of the relationship between Nc and body weight.

Discussion

The present study has revealed correlations in primates of Nc, Ve and Se, with TS, HRt, and HRi. Re-

sults of analyses using any of these indices of ECIs were similar, and the ECIs were positively and most closely correlated with TS in the prosimians, with HRt in the New World monkeys, and with HRi in the Old World monkeys. Furthermore, the correlations between ECIs and the ecological variables were usually stronger than the correlation between brain weight or body weight and the same ecological variables. Since ECIs are indicative of the size of cerebral cortical parts associated with advanced functions [Jerison, 1955, 1973; Hofman, 1982], and are closely related to learning ability or intelligence [Riddell et al., 1976; Riddell and Corl, 1977; Riddell, 1979; Ohta et al., 1984, 1985], these results suggest that the cerebral cortical expansion may be associated with TS in prosimians, with HRt in New World monkeys, and with HRi in Old World monkeys.

It should be noted that correlational studies such as the present study have problems with respect to the testing of causal relationships. It is unclear whether a change in the value of an ecological variable, which is closely and positively correlated with the ECIs, is the cause or the effect of the change in values of ECIs. It may be the case that the increase in value of the ecological variable is both cause and effect of the increase in values of the ECIs. There may be dynamic interrelationships between a given ecological variable and cortical expansion: the increase in value of the ecological variable may generate selection pressures on the development of the cerebral cortex, which, in turn, may produce the increase in value of the ecological variable. Nevertheless, the present study serves to provide working hypotheses that different ecological factors are associated with cortical expansion in different taxonomic groups of primates, and that cerebral cortical areas have enlarged differentially among different taxonomic groups of primates.

Different Ecological Factors Associated with the Expansion of the Cerebral Cortex in Different Taxonomic Groups

With respect to correlations of TS, HRt, and HRi with ECIs, different ecological factors appear to be operative in cortical expansion among the different taxonomic groups of primates. In the prosimians examined, the TS was positively and closely correlated with ECIs, so the development of socialization or social interaction may have been one of the critical factors in the expansion of the cerebral cortex in prosimians. A similar hypothesis was proposed by Jolly [1966, 1972]. In the New World monkeys examined, the size of the HRt was positively and closely correlated with ECIs. It is, therefore, possible that expansion of HRt may have been one of the critical factors in the development of the cerebral cortex of New World monkeys. The expansion of the size of home range may have been associated with the development of cerebral cortical areas involved in mnemonic, spatial, and motoric abilities which are more critical for survival in home ranges of a large area. In Old World monkeys, the size of HRi may have been one of the critical factors in the development of cerebral cortical areas. Since HRi in Old World monkeys is closely related to habitat quality, represented by the type of vegetation and the degree to which the habitat is disturbed [Takasaki, 1981, 1984], it follows logically that the cortical expansion in Old World monkeys may have been associated with changes of the quality of habitat. Among Old World monkeys, those living in poorer habitats tend to have wider HRi [Takasaki, 1981, 1984], and, in the present study, the terrestrial species showed wider HRi and larger Nc than the arboreal species. It is, therefore, possible that the deterioration in the quality of the habitat, accompanied by the move from arboreal to terrestrial habitats, might have been one of the critical factors in the development of the cerebral cortex of Old World monkeys, although further sophisticated analyses are necessary to confirm this hypothesis. Thus, cortical expansion may have been associated with the development of social interaction in prosimians, with the expansion of the home range in New World monkeys, and with the deterioration in the quality of the habitat in Old World monkeys.

Differential Cortical Growth among Different Taxonomic Groups of Primates

It is well known that different cortical areas are involved in different functions. Since different ecological factors may be involved in different cortical functions, it is possible that the cortical areas have grown differentially among the different taxonomic groups of primates. For example, temporal cortical areas are responsible for social interactions [Franzen and Myers, 1973] and are involved in recognition of faces [Perrett et al., 1979, 1982; Desimone et al., 1984], which may be critical for social interactions. These temporal cortical areas, therefore, may have devel-

oped in relation to the increase in the sizes of troops of prosimians. The parietal cortical areas are closely associated with spatial and motoric abilities [for a review, see Hyvärinen, 1982], and the temporal cortical areas are involved in visual mnemonic ability [Mishkin, 1982]. Since the expansion of the home range may be associated with the development of these abilities, it is possible that these parietal and temporal cortical areas may have developed with the expansion of the home ranges of New World monkeys. Indeed, it appears that New World monkeys with larger home ranges tend to show wider parietal and temporal cortical areas [Sawaguchi et al., unpubl. observation]. In the case of the Old World monkeys, since purposive, organized, and active behaviors would be critical for increasing fitness in relatively poor habitats, and since the prefrontal and premotor areas are responsible for such behaviors [Moll and Kuypers, 1977; Kubota, 1978; Fuster, 1981], these cortical areas may have developed in relatively poor habitats such as the savanna. Indeed, Watanabe et al. [1987] have reported that baboons, which live in the savanna, have a relatively wider premotor area than macaque monkeys whose HRi are much smaller than those of baboons. Thus, cortical areas, being associated with different ecological factors, may have developed differentially among different taxonomic groups of primates. The different patterns of cortical sulci and convolution between New World and Old World monkeys [Falk, 1980] may support this hypothesis, and systematic morphometric studies of cortical areas in members of different axonomic groups of primates will provide further insight into this problem.

Conclusion

The cerebral cortex may have developed in response to different ecological factors among different taxonomic groups of primates, and the cerebral cortical areas may have developed differentially among the groups of primates. Therefore, although it has been suggested that different axonomic groups of living primates represent different phases of a 'progressive' course of primate evolution towards Hominidae [for reviews, see Le Gros Clark, 1970, 1971], it is likely that the cerebral cortex has been differentially developed in different groups of primates as the result of different selective pressures during the course of primate evolution.

The present study does not exclude the possibility that there may be critical factors for the development of the cerebral cortex in primates other than the ecological factors examined here. Further comparative studies, similar to the present study, and further morphometric, neuroanatomical, psychological, and ecological studies may extend the results of the present study and help to reveal the nature of the evolution of the cerebral cortex in primates.

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Appendix

Estimates of ecological variables, body weight, and brain weight for the species included in the analysis

	TS	HRt ha	HRi ha	P g	E g
Prosimians					
Lemuroidea					
Lemur catta	18.0	7.4	0.41	2,700	25.6
Lemur fulvus	9.5	0.9	1.41	2,200	25.2
Lemur mongoz	2.5	1.2	0.46	1,800	21.8
Lepilemur mustelinus	1	0.2	0.2	625	9.5
Microcebus murinus	1	0.2	0.2	85	1.8
Indri indri	3	23	7.67	10,500	34.5
Propithecus verreauxi	6.5	7.6	1.17	3,600	27.5

Appendix (continued)

	TS	HRt ha	HRi ha	P g	E g
Lorisoidea					
Loris tardigradus	1	1	1	275	6.7
Perodicticus potto	1	10	10	1,050	14.3
Galago demidovii	1	1	1	62.5	2.7
Galago crassicaudatus	1	6	6	1,340	11.8
Galago alleni	1	4	4	250	6.1
New World monkeys					
Ceboidea					
Cebus capucinus	10	86	8.6	3,250	80.8
Cebus apella	10	80	8.0	2,480	71.0
Saimiri oerstedii	25	18	0.7	665	25.7
Alouatta seniculus	8.5	4.9	0.58	7,250	57.9
Callicebus moloch	3	4.2	1.4	1,075	19.0
Callicebus torquatus	3.5	20	5.7	1,100	22.4
Ateles geoffroyi	12	60	5.0	6,600	110.9
Ateles belzebuth	23	320	14	6,000	143.1
Old World monkeys					
Cercopithecoidea					
Cercocebus galeritus	19	127.5	6.7	7,850	114.7
Cercocebus albigena	15	410	27.3	7,700	99.1
Cercopithecus mitis	13.5	14	1.0	6,000	75.0
Cercopithecus cephus	10	35	3.5	3,500	63.6
Cercopithecus nictitaus	20	67	3.4	5,400	78.6
Cercopithecus pogonias	15	78	3.2	3,750	71.1
Cercopithecus neglectus	5	15	3.0	5,480	70.8
Cercopithecus aethiops	15	27	1.8	4,200	59.8
Micopithecus talapoin	70	120	1.7	1,250	37.7
Erythrocebus patas	20	5,200	260	7,800	106.6
Macaca mulatta	85	1,500	17.6	4,500	95.1
Macaca fascicularis	23	32	1.4	5,000	69.2
Macaca nemestrina	35	200	5.7	9,100	106.6
Macaca radiata	35	520	9.0	5,150	76.8
Macaca sinica	12.5	37.5	3.0	4,950	69.9
Macaca fuscata	47.5	575	12.1	10,400	109.1
Papio anubis	34	2,430	71.5	16,500	175.1
Papio cynocephalus	43	1,670	38.8	17,500	169.1
Colobus guereza	11	15	1.4	10,500	82.3
Colobus badius	41	67	1.6	8,150	73.8
Presbytis cristatus	32	20	0.6	8,350	64.0
Presbytis johnii	13	23	1.8	13,400	84.6
Presbytis melalophos	9.3	21	2.3	6,650	80.0
Presbytis obscurus	10	29	2.9	7,400	67.6
Presbytis entellus	19	340	17.9	14,900	135.2
Presbytis senex	8	12	1.5	8,150	64.9
Nasalis larvatus	20	130	6.5	15,100	94.2

Values for troop size and troop home range were taken from several field studies cited by Milton and May [1976] and/or Clutton-Brock and Harvey [1977a, b]. The values for body and brain weight are the means for those of female and male adults and were obtained from the systematic data of Harvey and Clutton-Brock [1985]. TS = Troop size; HRt = troop home range in hectares (ha); HRi = individual home range in hectares; P = body weight in grams (g); E = brain weight in grams.

References

Allman, J.: Reconstructing the evolution of the brain in primates through the use of comparative neurophysiological and neuroanatomical data; in Armstrong, Falk, Primate brain evolution, pp. 13–28 (Plenum Press, New York 1982).

Baker, J.; Peterson, S.E.; Newsome, W.T.; Allman, J.M.: Visual response properties of neurons in four extrastriate visual areas of the owl monkey *(Aotus trivigatus)*: a quantitative comparison of medial, dorsomedial, dorsolateral, and middle temporal areas. J. Neurophysiol. *45:* 397–416 (1981).

Brody, E.G.; Rosvold, H.E.: Influences of prefrontal lobectomy on social interaction in a monkey group. Psychosom. Med. *14:* 406–415 (1952).

Carlson, M.; Huerta, M.F.; Cusick, C.G.; Kaas, J.H.: Studies on the evolution of multiple somatosensory representations in primates: the organization of anterior parietal cortex in the New World Callitrichid, *Saguinus*. J. comp. Neurol. *246:* 409–426 (1986).

Cheney, D.; Robert, S.; Smuts, B.: Social relationships and social cognition in nonhuman primates. Science *234:* 1361–1366 (1986).

Clutton-Brock, T.H.; Harvey, P.H.: Primate ecology and social organization. J. Zool., Lond. *183:* 1–39 (1977a).

Clutton-Brock, T.H.; Harvey, P.H.: Species differences in feeding and ranging behaviour in primates; in Clutton-Brock, Primate ecology, pp. 557–584 (Academic Press, London 1977b).

Clutton-Brock, T.H.; Harvey, P.H.: Primates, brain and ecology. J. Zool., Lond. *190:* 309–323 (1980).

Desimone, R.; Albright, T.D.; Gross, C.G.; Bruce, C.: Stimulus-selective properties of inferior temporal neurons in the macaque. J. Neurosci. *4:* 2051–2062 (1984).

Falk, D.: Comparative study of the endocasts of New and Old World monkeys; in Ciochon, Chiareli, Evolution of the New World monkeys and continental drift, pp. 275–292 (Plenum Press, New York 1980).

Franzen, E.A.; Myers, R.E.: Neural control of social behavior: prefrontal and anterior temporal cortex. Neuropsychology *11:* 141–157 (1973).

Fuster, J.M.: Prefrontal cortex in motor control; in Brooks, Handbook of physiology. Nervous system, vol. 2: Motor control. Part II, pp. 1149–1178 (American Physiological Society, Washington 1981).

Gross, C.G.: Inferotemporal cortex and vision; in Stellar, Sprague, Progress in physiological psychology, vol. 5, pp. 77–123 (Academic Press, New York 1973).

Harvey, P.H.; Clutton-Brock, T.H.: Life history variation in primates. Evolution *39:* 559–581 (1985).

Harvey, P.H.; Clutton-Brock, T.H.; Mace, G.M.: Brain size and ecology in small mammals and primates. Proc. natn. Acad. Sci. USA *77:* 4387–4389 (1980).

Hofman, M.A.: Encephalization in mammals in relation to the size of the cerebral cortex. Brain Behav. Evol. *20:* 84–96 (1982).

Holloway, R.L.: Cranical capacity, neural reorganization, and hominid evolution: a search for more suitable parameters. Am. J. phys. Anthrop. *25:* 305 (1966).

Holloway, R.L.: On the meaning of brain size. Science *184:* 677–679 (1974).

Humphrey, N.K.: The social function of intellect; in Baterson, Hinde, Growing points in ecology, pp. 303–317 (Cambridge University Press, London 1976).

Hyvärinen, J.: The parietal cortex of monkey and man (Springer, Berlin 1982).

Jerison, H.J.: Brain to body ratios and the evolution of intelligence. Science *121:* 447–449 (1955).

Jerison, H.J.: Evolution of the brain and intelligence (Academic Press, New York 1973).

Jerison, H.J.: Animal intelligence as encephalization. Phil. Trans. R. Soc. *308:* 21–35 (1985).

Jolly, A.: Lemur social behavior and primate intelligence. Science *153:* 501–506 (1966).

Jolly, A.: The evolution of primate behavior (MacMillan, New York 1972).

Kaas, J.H.: What, if anything is SI? Organization of the first somatosensory area of cortex. Physiol. Rev. *63:* 206–231 (1983).

Kubota, K.: Neuron activity in the dorsolateral prefrontal cortex of the monkey and initiation of behavior; in Ito, Tsukahara, Kubota, Yagi, Integrative control functions of the brain, vol. 1, pp. 407–417 (Kodansha-Elsevier, Tokyo 1978).

Le Gros Clark, W.E.: History of the primates: an introduction to study of fossil man (British Museum, London 1970).

Le Gros Clark, W.E.: The antecedents of man (Edinburgh University Press, Edinburgh 1971).

Milton, K.; May, M.L.: Body weight, diet and home range area in primates. Nature *259:* 459–462 (1976).

Mishkin, M.: A memory system in the monkey. Phil. Trans. R. Soc. *298:* 85–95 (1982).

Moll, L.; Kuypers, H.G.J.M.: Premotor cortical ablations in monkeys: contralateral changes in visually guided reaching behavior. Science *198:* 317–319 (1977).

Ohta, H.; Ishida, H.; Matano, S.: Learning set formation in ring-tailed lemur *(Lemur catta)*. Folia primatol. *43:* 53–58 (1984).

Ohta, H.; Matsutani, S.; Ishida, H.; Matano, S.: Learning set formation in common tree shrew *(Tupaia glis)*. Folia primatol. *44:* 204–209 (1985).

Overall, J.E.; Klett, C.J.: Applied multivariate analysis (McGraw-Hill, New York 1972).

Penfield, W.: Speech, perception and uncommitted cortex; in Eccles, Brain and conscious experience, pp. 217–237 (Springer, Berlin 1966).

Perrett, D.I.; Rolls, E.T.; Caan, W.: Temporal lobe cells of the monkey with visual responses selective for face. Neurosci. Lett. *53:* 358 (1979).

Perrett, D.I.; Rolls, E.T; Caan, W.: Visual neurons responsive to faces in the monkey temporal cortex. Exp. Brain Res. *47:* 329–342 (1982).

Riddell, W.I.: Cerebral indices and behavioral differences; in Hahn, Jensen, Dudek, Development and evolution of brain size: behavioral implications, pp. 89–109 (Academic Press, London 1979).

Riddell, W.I.; Corl, K.: Comparative investigation of the relationship between cerebral indices and learning abilities. Brain Behav. Evol. *14:* 385–398 (1977).

Riddell, W.I.; Gravetter, F.; Rogers, W.: Further investigation of the relationship between brain indices and learning. Physiol. Behav. *17:* 231–234 (1976).

Sanides, F.: Functional architecture of motor and sensory cortices

in primates in light of a new concept of neocortical evolution; in Noback, Montagna, The primate brain, pp. 137–208 (Appleton Century Crofts, New York 1970).

Takasaki, H.: Troop size, habitat quality, and home range area in Japanese macaques. Behav. Ecol. Sociobiol. *9:* 277–281 (1981).

Takasaki, H.: A model for relating troop size and home range area in a primate species. Primates *25:* 22–27 (1984).

Van Essen, D.C.: Visual areas of the mammalian cerebral cortex. Annu. Rev. Neurosci. *2:* 227–263 (1979).

Watanabe, K.; Arikuni, T.; Kubota, K.: Studies of structures and functions of the frontal lobe of the baboon. Part 1. Primate Res. *4:* 15 (1987).

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